

for electric vehicle (EV) power modules, radio frequency (RF) hardware, automotive electronic control units (ECUs), power semiconductors, medical electronics, aerospace systems, defence equipment, and telecom infrastructure.

Use case: From aerospace to EVs

Originally developed for aerospace and defence applications, vacuum reflow is increasingly being adopted for high-value assemblies where joint consistency and reliability are critical. By reducing defects early in production, vacuum reflow helps streamline manufacturing, lowers rework, improves inspection yields, and increases process consistency.

The process combines precise thermal profiling with carefully controlled vacuum cycles, often using inert gases such as nitrogen to minimise oxidation and ensure proper solder wetting. Modern systems support a wide range of packages, including BGAs, QFNs, DFNs, IGBTs, power modules, and other fine-pitch components. Real-time monitoring of oxygen concentration, temperature profiles, vacuum pressure, and process data further helps manufacturers maintain consistency while meeting IPC-A-610 and MIL-STD quality requirements.

Performance and industry outlook

Both batch and inline reflow systems can accommodate complex assemblies, while lower oxygen levels and careful flux control contribute to more uniform and robust solder joints.

To support these applications, several manufacturers have

Manufacturer	Website	Headquarters
BTU International, Inc.	www.btu.com	USA
Heller Industries, Inc.	www.hellerindustries.com	USA
Kurtz Ersä GmbH	www.kurtzersa.com	Germany
Rehm Thermal Systems GmbH	www.rehm-group.com	Germany
Shenzhen JT Automation Equipment Co., Ltd	www.jt-int.com	China
SMT Wertheim GmbH	www.smt-wertheim.de	Germany
Suneast Technology Int'l Ltd	www.suneast1984.com	China
Tamura Corporation	www.tamuracorp.com	Japan

Supplier/Distributor	Website	Headquarters
ETA International Pte Ltd (I.C.T. Partner)	www.etasmt.com.sg	Bengaluru
Heller India (Heller Industries Rep)	www.hellerindustries.in	Chennai
Paar Automation	www.paarautomation.co.in	Hosur

developed specialised vacuum reflow systems with advanced chamber designs, process monitoring, and flux management capabilities. One example is TSM Co., Ltd's TRV III vacuum reflow oven, which incorporates features such as a CAP-type vacuum chamber, finger-hook PCB transport, real-time oxygen and temperature monitoring, and an integrated flux management system to improve process consistency and reduce solder voiding.

"Traditional solder processes often struggle to keep voids consistently low," says Phil Zarrow from Indium Corporation. "Vacuum reflow can push void rates towards single-digit levels, enabling stronger, more reliable solder joints."

Depending on the solder alloy, package design, and process optimisation, vacuum reflow can reduce voiding by a factor of six to nine compared with conventional

nitrogen reflow. Well-optimised processes can achieve void levels below 1% in many applications.

"Vacuum reflow is the next step in achieving a reliable reflow process for today's miniature electronic assemblies," says Narayan Kumar Jayaraman, Managing Director, Sanjay Technologies.

As circuit miniaturisation, process automation, and Industry 5.0 practices continue to advance, vacuum reflow is becoming an increasingly important manufacturing process because of its consistent quality and compatibility with modern production lines. Its ability to improve reliability, thermal performance, and process consistency is contributing to wider adoption across automotive, telecom, industrial power electronics, and other high-performance applications. Market forecasts indicate continued growth, particularly across automotive, telecom, and high-power industrial applications. **EFY**

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Saba Aafreen, Tech Journalist at EFY, blends on-ground industrial experience with a growing focus on AI-driven technologies in the evolving electronics industry.